

PRE-REGISTRATION P9

H3K27me3 Enrichment at Neural Crest Migration Loci in Domestic *Sus scrofa* vs. Wild Boar Embryos: The ChIP-seq Experiment

EZH2-Mediated Epigenetic Silencing of the Neural Crest Migration Program
as the Molecular Mechanism of Field-Specified Developmental Arrest
Predicted and Locked Before Any Sequencing Data Exists

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THE PRE-REGISTERED CLAIM

The molecular mechanism maintaining the domestication developmental arrest is EZH2-mediated H3K27me3 deposition at neural crest migration driver loci during embryogenesis. This is the same mechanism that maintains cancer attractor arrest via EZH2-mediated silencing of the luminal identity program in triple-negative breast cancer.

This pre-registration specifies: the loci to be measured, the developmental stage at which the measurement must be taken, the comparison to be performed, the predicted direction of enrichment, the scoring criteria, the control conditions, and the success criteria — all locked before any ChIP-seq data is generated.

Primary prediction (P9): Domestic pig embryos at gestational day E14–E18 will show significantly elevated H3K27me3 enrichment at the following neural crest migration driver loci relative to wild boar embryos at the same developmental stage with equivalent total H3K27me3 signal: *BMP4*, *BMP7*, *WNT1*, *WNT3A*, *CXCL12*, *CXCR4*, *EDNRB*, *SOX10*, *SNAI2*. Minimum 4 of 9 target loci reach enrichment ratio ≥ 1.5 (domestic/wild boar) at $q < 0.05$ (FDR-corrected).

1. Background and Theoretical Basis

1.1. The Open Causal Question

Wilkins, Wrangham and Fitch (2014) identified that all domestication syndrome traits map to a single developmental perturbation: mild reduction in neural crest cell (NCC) migration during embryogenesis. They attributed this reduction to genetic change — alleles affecting NCC behaviour co-selected alongside tameness alleles.

The field-arrest model (Lawson, 2026, [doi:10.5281/zenodo.19094935](https://doi.org/10.5281/zenodo.19094935)) proposes that the causal origin is not genetic but epigenetic: the domestication coherence gradient field (N_{domestic}) suppresses the maternal HPA axis, which alters the intrauterine coherence gradient environment, which establishes H3K27me3 enrichment at NCC migration driver loci via EZH2 activity, which arrests NCC migration at the juvenile-proximate developmental stage.

The molecular mechanism proposed is EZH2-mediated H3K27me3 silencing of the NCC migration program. This is the same molecular mechanism that maintains cancer attractor arrest in triple-negative breast cancer (TNBC), where EZH2 deposits H3K27me3 at *FOXA1*, *GATA3*, and *ESR1* promoters, silencing the luminal identity program and maintaining the cancer attractor basin.

The primary field-level experimental prediction of the field-arrest model is pre-registered as P7 ([doi:10.5281/zenodo.19096128](https://doi.org/10.5281/zenodo.19096128)).

P9 is the molecular mechanism pre-registration: it specifies the epigenetic signature that must be present if the field-arrest mechanism is correct, independently of whether P7 has yet been experimentally tested. If P9 confirms, the molecular mechanism of domestication and the molecular mechanism of cancer are the same mechanism operating at different scales:

	Cancer (TNBC)	Domestication (<i>Sus scrofa</i>)
Scale	Single cell	Whole organism
Silenced program	Luminal identity (<i>FOXA1</i> , <i>GATA3</i> , <i>ESR1</i>)	NCC migration program (<i>BMP4/7</i> , <i>WNT1/3A</i> , <i>CXCL12/CXCR4</i> , <i>ED-NRB</i> , <i>SOX10</i> , <i>SNAI2</i>)
Silencing mechanism	EZH2 → H3K27me3	EZH2 → H3K27me3 (predicted)
Reversal agent	Tazemetostat	Tazemetostat (predicted — P11)
Arrest phenotype	Cancer	Domestic pig
Resumed phenotype	Luminal differentiation	Wild boar morphology

1.2. Why the NCC Migration Stage E14–E18 Is the Critical Window

Neural crest cell migration in *Sus scrofa* follows the standard vertebrate timeline: NCC specification at the dorsal neural tube begins at approximately E8–E10; delamination and active migration at E10–E14; differentiation into target tissues at E14–E20. The E14–E18 window represents the period of maximum active migration, when the gradient guidance environment (BMP, Wnt, CXCL12/CXCR4) is most critical for migration trajectory and therefore most sensitive to epigenetic modification of the guidance genes. H3K27me3 enrichment at migration driver loci at E14–E18 would silence the gradient guidance signals during their peak activity window, producing partial migration arrest at the juvenile-proximate stage without complete ablation (which would produce lethal NCC deficiency rather than viable domestic phenotype). The E14–E18 window is therefore the predicted critical measurement window. Earlier (E8–E12) would capture specification but not guidance arrest. Later (E20+) would capture post-differentiation states rather than the migration arrest itself.

1.3. Why This Matters Beyond Domestication

- If H3K27me3 at NCC migration loci in domestic embryos is confirmed, this establishes:
1. The molecular mechanism of domestication is epigenetic, not genetic. The genome was never modified. The epigenome was.
 2. The same enzyme (EZH2) that maintains cancer attractor arrest at the single-cell scale maintains domestication developmental arrest at the organismal scale. Scale-invariant epigenetic arrest mechanism confirmed.
 3. Tazemetostat — currently in clinical use for EZH2-mutant cancers — is the predicted reversal agent for domestication arrest (P11). P9 confirmation creates the biological justification for P11.
 4. The field (*N*) → epigenome → developmental trajectory causal chain is confirmed at the molecular level. The attractor geometry framework ($S + N + G \rightarrow R$) is grounded in a specific, testable molecular mechanism.

2. The Pre-Registered Locus Set

2.1. Primary Target Loci

The following loci are pre-registered as the primary measurement targets. They were selected before literature consultation on *Sus scrofa* ChIP-seq data, based solely on their established roles in vertebrate NCC migration guidance.

Gene	NCC Function	Rationale for Inclusion
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<i>BMP4</i>	NCC migration guidance gradient — dorsoventral axis	BMP4 establishes the dorsoventral coherence gradient that directs NCC migration away from the neural tube. Silencing disrupts the primary spatial guidance signal.
<i>BMP7</i>	NCC migration guidance gradient — complementary to BMP4	BMP7 acts as a complementary gradient signal to BMP4 in the NCC migration environment. Co-enrichment at both BMP4 and BMP7 would indicate disruption of the complete BMP guidance axis.
<i>WNT1</i>	NCC specification and delamination trigger	WNT1 is required for NCC specification and initiates the delamination program. H3K27me3 enrichment here would arrest the program before migration begins.
<i>WNT3A</i>	NCC specification amplification and dorsal migration support	WNT3A amplifies WNT1 signalling in post-delamination NCC. Enrichment here reduces migration competence.
<i>CXCL12</i>	Chemokine gradient — directional migration guidance	CXCL12 (SDF-1) establishes the chemotactic gradient that directs NCC to target tissues. The CXCL12/CXCR4 axis is the primary directional guidance mechanism. Silencing of CXCL12 in source tissue eliminates the gradient.
<i>CXCR4</i>	Chemokine receptor on migrating NCC — gradient sensor	CXCR4 on NCC cells detects the CXCL12 gradient. H3K27me3 enrichment at CXCR4 reduces NCC responsiveness to the directional signal.
<i>EDNRB</i>	Endothelin receptor B — melanocyte and enteric NCC migration	EDNRB is specifically required for melanocyte precursor and enteric nervous system NCC migration. Enrichment here specifically explains pigmentation (depigmentation syndrome trait) and connects directly to Marcström's white patch observation.
<i>SOX10</i>	NCC identity maintenance transcription factor	SOX10 maintains multipotency and migration competence of NCC during migration. H3K27me3 at SOX10 would cause NCC to exit the migration-competent state prematurely.
<i>SNAI2</i>	Epithelial-to-mesenchymal transition (EMT) factor for NCC delamination	SNAI2 (Slug) drives the EMT that allows NCC to delaminate from the neural tube and begin migration. H3K27me3 enrichment here arrests delamination.

2.2. Secondary Target Loci

The following loci are pre-registered as secondary targets. Enrichment here would extend the prediction beyond NCC migration to broader developmental programs. They are not included in the primary success criterion but are reported as exploratory:

Gene	Rationale
<i>TWIST1</i>	NCC EMT regulator, craniofacial development. H3K27me3 here would explain snout shortening.
<i>FOXD3</i>	NCC multipotency maintenance. Early NCC identity factor.
<i>PAX3</i>	NCC specification at the neural plate border. Upstream of SOX10.
<i>TFAP2A</i>	NCC specification and survival during migration.
<i>DLX2</i>	Craniofacial NCC differentiation. H3K27me3 here would explain skull shape changes.
<i>PHOX2B</i>	Autonomic nervous system NCC derivation. Adrenal chromaffin cell specification.

2.3. Negative Control Loci

The following loci are pre-registered as negative controls. H3K27me3 enrichment is *not* predicted at these loci. If enrichment is found at negative control loci at comparable levels to positive target loci, this indicates

a global increase in H3K27me3 rather than locus-specific enrichment, which would require re-analysis with stricter normalisation.

Gene	Rationale for Selection as Negative Control
<i>GAPDH</i>	Housekeeping gene. Should not be H3K27me3 enriched in any active cell type.
<i>ACTB</i>	Housekeeping gene. Standard negative control for histone modification ChIP.
<i>MYF5</i>	Skeletal muscle specification factor. Not expressed in NCC or relevant to NCC development. Should not be enriched.
<i>HBB</i>	Haemoglobin beta. Erythroid lineage marker. Not expressed in embryonic NCC tissue.

3. Experimental Design

3.1. Overview

H3K27me3 ChIP-seq on embryonic neural tube and adjacent embryonic tissue (NCC migration zone) from domestic *Sus scrofa* (commercial breed) and wild boar (*Sus scrofa* wild-caught, confirmed no domestic admixture) at gestational day E14–E18. Comparison: H3K27me3 enrichment at pre-registered NCC migration driver loci, domestic vs. wild, same developmental stage, same tissue type.

No genetic modification. No experimental intervention on the animals. Pure comparison of epigenetic state at equivalent developmental stage in domestic vs. wild genome-matched tissue.

3.2. Animal Subjects

Domestic embryos: Commercial breed pregnant sows (Large White or Landrace, confirmed breed registration). Embryos collected at confirmed E14, E16, and E18 (timed from confirmed mating). Minimum $n = 3$ litters per timepoint. Minimum $n = 3$ embryos per litter (one per litter used for ChIP; remaining for RNA-seq validation — see Section 3.4).

Wild boar embryos: Wild-caught pregnant *Sus scrofa* sows, confirmed by SNP panel to have $< 5\%$ domestic admixture (same standard as P7, [doi:10.5281/zenodo.19096128](https://doi.org/10.5281/zenodo.19096128)). Gestational age confirmed by crown-rump length measurement against established *Sus scrofa* growth curves. Embryos collected at equivalent E14, E16, and E18. Minimum $n = 3$ litters per timepoint.

Tissue collection: Embryos collected immediately post-euthanasia of dam (licensed, IACUC-approved procedure). Dissection within 30 minutes of collection. Neural tube region (dorsal cranial neural tube + dorsal trunk neural tube, E14–E18 stage) dissected on ice, snap-frozen in liquid nitrogen within 60 minutes of collection. Parallel tissue for RNA-seq fixed in RNAlater.

3.3. ChIP-seq Protocol

3.3.1. Chromatin Preparation

Tissue (neural tube + adjacent NCC migration zone, approximately 50–100 mg per sample) cross-linked with 1% formaldehyde, 10 minutes at room temperature. Quenched with glycine. Nuclei isolated by Dounce homogenisation and differential centrifugation. Chromatin sheared to 200–500 bp by sonication (Bioruptor Pico or equivalent: 30 cycles, 30 s on/30 s off at 4°C). Shear size confirmed by Bioanalyzer before proceeding.

3.3.2. Immunoprecipitation

Anti-H3K27me3 antibody: Cell Signaling Technology #9733 (rabbit monoclonal C36B11 — the validated antibody used across the domestication syndrome field’s epigenetic studies and in all referenced cancer ChIP-seq work) or Abcam ab192985 (rabbit monoclonal). Both antibodies must be tested in parallel if sufficient material allows.

Negative IP control: rabbit IgG isotype control (same species and format as primary antibody, same concentration).

Positive IP control: H3K27me3 at *CDKN2A* (p16/INK4a locus) — a validated PRC2/EZH2 target enriched by H3K27me3 in multiple species and cell types. Must confirm > 10 -fold enrichment over IgG at *CDKN2A* before any sample is considered valid.

ChIP input: 5% of chromatin retained as input control per sample. Library preparation: NEBNext Ultra II ChIP-seq Library Prep Kit or equivalent. Sequencing: Illumina paired-end 150 bp, minimum 30 million reads per sample.

3.3.3. Sequencing Quality and Alignment

Quality control: FastQC on all raw reads before alignment. Samples with < 25 million mappable reads or > 30% duplication rate excluded and resequenced.

Alignment: BWA-MEM2 to the *Sus scrofa* reference genome (Sscrofa11.1, Ensembl release 109 or equivalent current release). Duplicate removal: Picard MarkDuplicates. Blacklist regions excluded using ENCODE *Sus scrofa* blacklist (or ENCODE porcine blacklist if available; otherwise mammalian consensus blacklist lifted over to Sscrofa11.1).

3.3.4. Peak Calling and Differential Analysis

Peak calling: MACS2 (–broad flag for H3K27me3 broad domain marks, *q*-value threshold 0.05). Input subtracted per sample.

Differential enrichment at pre-registered loci: DESeq2 on read counts in ±5kb windows centred on the transcription start site (TSS) of each target gene, plus any MACS2-called peaks within 50 kb of TSS.

Comparison: domestic (all timepoints) vs. wild boar (all timepoints), with gestational day as covariate. Primary comparison: E16 (mid-window) domestic vs. E16 wild boar.

FDR correction: Benjamini-Hochberg across all pre-registered loci in a single correction set. *q* < 0.05 threshold. Enrichment ratio threshold: ≥ 1.5 (domestic/wild boar) at the TSS±5 kb window.

3.4. RNA-seq Validation

For each embryo where ChIP material is collected, parallel RNA-seq is performed on adjacent neural tube tissue from the same dissection.

Rationale: H3K27me3 enrichment at a locus should correspond to reduced transcription of that gene. RNA-seq confirms that the predicted H3K27me3 enrichment produces the predicted transcriptional silencing. If H3K27me3 is elevated at a locus but transcription is unchanged, the enrichment is not functionally relevant.

Protocol: Total RNA extraction: RNeasy Micro Kit (Qiagen) or equivalent. RNA quality: RIN ≥ 7.0 required (Bioanalyzer). Library: NEBNext Ultra II RNA-seq with poly-A selection. Sequencing: Illumina paired-end 150 bp, minimum 20 million reads per sample. Alignment: STAR to Sscrofa11.1. Differential expression: DESeq2.

Predicted direction: Genes with elevated H3K27me3 in domestic embryos (from ChIP-seq primary analysis) should show *reduced* expression in domestic embryo neural tube relative to wild boar at equivalent stage. Confirmed H3K27me3 enrichment + reduced expression = dual confirmation of silencing.

4. The Two Models and Their Diverging Predictions

Outcome	Genetic model	Field-arrest model
H3K27me3 at NCC loci in domestic vs. wild embryos (same developmental stage)	No difference predicted. Genetic differences affect allele-encoded programs, not H3K27me3 patterns at equivalent developmental stages.	Elevated in domestic embryos at ≥ 4/9 pre-registered loci. Enrichment ratio ≥ 1.5 at <i>q</i> < 0.05.
EZH2 protein level in domestic vs. wild embryo NCC zone	No specific prediction.	Elevated in domestic embryos at E14–E18 in the NCC migration zone. (Exploratory; not in primary success criterion.)
Expression of NCC migration drivers in domestic vs. wild (RNA-seq)	No difference predicted at equivalent developmental stages.	Reduced in domestic embryos at loci with elevated H3K27me3.
H3K27me3 at negative control loci	No enrichment predicted.	No enrichment predicted. (Same prediction — used only to detect global artefacts.)

The distinguishing comparison is: H3K27me3 enrichment at NCC migration driver loci in domestic vs. wild embryos at the same developmental stage with equivalent global H3K27me3 signal. This comparison is not confounded by developmental stage differences (same stage), by tissue differences (same dissection protocol), or by global chromatin accessibility differences (input-subtracted ChIP, housekeeping gene negative controls).

5. Success Criteria

5.1. Primary Success Criterion

P9-SUCCESS requires ALL of the following:

Criterion 1 (ChIP quality): > 10-fold H3K27me3 enrichment over IgG at the *CDKN2A* positive control locus confirmed in $\geq 80\%$ of samples before any target locus analysis proceeds.

Criterion 2 (NCC locus enrichment): ≥ 4 of the 9 pre-registered primary target loci show H3K27me3 enrichment ratio (domestic/wild boar) ≥ 1.5 at $q < 0.05$ (BH-FDR corrected across all 9 loci) at the E16 primary timepoint.

Criterion 3 (Direction specificity): Mean H3K27me3 enrichment at negative control loci (GAPDH, ACTB, MYF5, HBB) does not differ significantly between domestic and wild boar samples ($p > 0.05$, confirming the enrichment at target loci is locus-specific, not a global H3K27me3 increase).

Criterion 4 (RNA-seq concordance): ≥ 3 of the loci meeting Criterion 2 also show significantly reduced mRNA expression in domestic embryo neural tube relative to wild boar at the same stage ($q < 0.05$, DESeq2), confirming that H3K27me3 enrichment corresponds to functional silencing.

5.2. Graded Confirmation Levels

Outcome	Interpretation
All 4 criteria met	Full confirmation. EZH2/H3K27me3 silencing of the NCC migration program is the molecular mechanism of domestication developmental arrest. Proceed immediately to P11 (tazemetostat experiment). Publish P9 result and update theory paper with molecular mechanism confirmed.
Criteria 1–3 met, Criterion 4 not met	Structural confirmation, functional confirmation absent. H3K27me3 enrichment is present at NCC loci but not reducing transcription. Possible explanation: the enrichment affects enhancer elements not captured by TSS $\pm$ 5 kb window. Extend to enhancer-focused analysis (H3K27ac ChIP-seq to map active enhancers in wild boar; confirm their suppression in domestic). Still considered partial confirmation.
Criterion 2 met (3/9 loci only, below threshold)	Trend toward confirmation, below pre-specified threshold. Increase sample size to $n \geq 5$ litters per timepoint and repeat. If 3/9 becomes 4/9 at larger n : full analysis. If remains $< 4/9$: failure diagnosis.
Criterion 1 met, Criterion 2 not met	True negative. ChIP quality confirmed (positive control passed). H3K27me3 enrichment at NCC loci is absent. EZH2/H3K27me3 is not the mechanism. Failure diagnosis applied.

6. Control Conditions and Confound Management

6.1. Critical Controls

C1 — Developmental stage matching (primary confound): H3K27me3 patterns change substantially across developmental stages. Domestic and wild boar embryos must be confirmed at equivalent developmental stage by *independent* criteria beyond gestational day: crown-rump length measurement against published *Sus scrofa* staging tables, and somite count where available. Any sample pair with > 1 day estimated developmental age difference is excluded from primary analysis and reported separately.

C2 — Tissue dissection equivalence: The NCC migration zone is a small, specific region of the embryo. Contamination by non-NCC tissue (e.g., epidermis, notochord, somites) will dilute the signal. Dissection must be performed by the same trained dissector for all samples. Dissection protocol must be validated on pilot samples by H&E histology of adjacent sections to confirm correct tissue composition.

C3 — Antibody specificity: Two independent anti-H3K27me3 antibodies must be used in parallel (CST #9733 and Abcam ab192985). Loci called as enriched must show enrichment with both antibodies. Any locus enriched with only one antibody is reported as exploratory only.

C4 — Input normalisation: 5% input control retained per sample. All enrichment ratios computed after input subtraction. Samples with input library quality < 15 million mappable reads excluded.

C5 — Domestic admixture exclusion: Wild boar samples must be confirmed at < 5% domestic admixture by SNP panel (same standard as P7, [doi:10.5281/zenodo.19096128](https://doi.org/10.5281/zenodo.19096128)). Any wild boar sample with ≥ 5% admixture excluded. This controls for the possibility that wild boar samples with partial domestic ancestry show partial H3K27me3 enrichment that contaminates the wild boar baseline.

C6 — Cross-laboratory replication: Primary analysis performed at one sequencing facility. Results submitted for replication at a second independent facility on a minimum of 3 embryo pairs (1 domestic, 1 wild boar, matched stage) before primary success criterion is declared.

6.2. Pre-specified Confound Interpretations

Confound: Breed-specific H3K27me3 differences unrelated to NCC. If all analysed loci (including negative controls) show elevated H3K27me3 in domestic vs. wild, this is a global methylation difference, not a locus-specific NCC arrest signal. The negative control criterion (Criterion 3) specifically detects this. If Criterion 3 fails, the dataset is invalid for the primary claim and requires reanalysis with global H3K27me3 normalisation (spike-in normalisation with *Drosophila* chromatin).

Confound: Different NCC developmental timing between domestic and wild embryos. If domestic pigs have an accelerated or delayed NCC migration timeline relative to wild boar (independent of the arrest), then E16 domestic and E16 wild boar may not be at equivalent NCC migration stages. This is detected by RNA-seq: the expression signature of NCC migration stage markers (*SOX10*, *SNAI2*, *FOXD3* expression in the neural tube) must be confirmed equivalent between domestic and wild boar samples included in the primary comparison.

7. Failure Diagnosis

If P9 fails (Criterion 1 passes, Criterion 2 does not), the following diagnoses are pre-specified:

Diagnosis F1 (Wrong histone mark): The arrest is maintained by a different Polycomb mechanism. H3K27me3 is the canonical PRC2 mark but PRC1 (H2AK119ub) or DNA methylation (5mCpG) could alternatively silence the NCC migration program. *Direction:* Perform H2AK119ub ChIP-seq and WGBS on the same tissue pairs. If H2AK119ub is enriched at NCC loci in domestic vs. wild boar, the mechanism is PRC1 rather than PRC2. The field-arrest model is confirmed; the specific enzyme is PRC1 rather than EZH2.

Diagnosis F2 (Wrong tissue: arrest occurs earlier): The epigenetic arrest is established at a pre-neural-crest stage (E8–E10, specification stage) rather than during active migration (E14–E18). H3K27me3 at migration guidance loci is secondary to the primary arrest elsewhere. *Direction:* Perform the same ChIP-seq protocol on E10–E12 embryos (specification stage) to test for earlier enrichment.

Diagnosis F3 (Wrong loci): The H3K27me3 arrest targets a different set of NCC-related genes not in the pre-registered panel. *Direction:* Perform genome-wide differential H3K27me3 analysis (not restricted to pre-registered loci). Identify which, if any, NCC-related loci show enrichment. Report as exploratory.

Diagnosis F4 (Mechanism is not epigenetic): The NCC migration reduction in domestic animals is genetic (allele-encoded), not epigenetic. The field does not establish H3K27me3. The genome specifies reduced NCC migration capacity directly. *Direction:* Perform whole-genome sequencing of domestic vs. wild boar embryos. Identify coding variants in the pre-registered NCC loci. If functional variants in

NCC migration genes are present and track with domestication, the genetic model is supported over the field-arrest model for this specific mechanism. This outcome would be published as a partial falsification of the epigenetic arm of the field-arrest model while leaving the basin-geometry framework intact.

8. Pathway to P11

P9 is the molecular prerequisite for P11 (tazemetostat in pregnant domestic sow, shifting F1 morphology toward wild-type). The logical dependency is:

P9 confirms H3K27me3 at NCC loci in domestic embryos	⇒	EZH2 is the arrest maintenance enzyme. Tazemetostat (EZH2 inhibitor) is the predicted reversal agent. P11 is justified and ready to proceed.
P9 fails (Diagnosis F1: PRC1/H2AK119ub instead)	⇒	Tazemetostat is the wrong drug. PRC1 inhibitor (e.g., PTC-596 targeting BMI1) is the predicted reversal agent. P11 is redesigned with PRC1 inhibitor.
P9 fails (Diagnosis F4: genetic mechanism)	⇒	P11 is not justified. No pharmacological reversal of the arrest is predicted. The field-arrest epigenetic model is falsified for this specific mechanism.

P9 is therefore not only a test of the field-arrest mechanism. It is the gateway experiment that determines the molecular target for P11 and, by extension, provides the biological justification for the cross-scale connection between domestication biology and EZH2 inhibitor cancer therapy.

9. Cost and Timeline

Item	Estimated Cost (USD)
Wild boar embryo collection (licensed, $n = 9$ litters, 3 per timepoint)	\$8,000–\$18,000
Domestic pig embryo collection ($n = 9$ litters, abattoir or farm)	\$3,000–\$6,000
Chromatin preparation and ChIP (2 antibodies $\times$ 2 groups $\times$ 3 timepoints $\times$ 3 replicates)	\$12,000–\$20,000
ChIP-seq library prep and sequencing (36 samples $\times$ 30M reads)	\$18,000–\$28,000
RNA-seq library prep and sequencing (36 samples $\times$ 20M reads)	\$10,000–\$16,000
Bioinformatic analysis (alignment, peak calling, DESeq2)	\$4,000–\$8,000
SNP genotyping for admixture confirmation	\$2,000–\$4,000
Cross-laboratory replication (3 embryo pairs)	\$8,000–\$12,000
Histological validation of tissue dissection	\$2,000–\$4,000
Total estimated cost	\$67,000–\$116,000

Timeline from ethical approval to primary data: 3–6 months (embryo collection is the rate-limiting step; sequencing and analysis: 6–8 weeks after collection).

Minimum timeline to replication: 8–10 months.

10. Reporting Standard

Results of this experiment will be reported against this pre-registration regardless of outcome. The following standards apply:

1. All 9 pre-registered primary target loci are reported. No selective reporting of only confirming loci.
2. All 4 negative control loci are reported. If any negative control locus shows enrichment at threshold, this is reported prominently, not buried.
3. The analysis is performed exactly as specified in Section 3.3. Any deviation must be declared in the methods with justification.

- 4. The cross-laboratory replication result must be reported in the same publication as the primary result, not deferred to a separate publication.
- 5. If P9 fails, the failure diagnosis and the follow-on direction are published in the primary report. A failure report is a result. It will be published.
- 6. Sequencing data deposited in GEO (Gene Expression Omnibus) before submission to journal, with accession number included in the methods.

11. Linked Records

The following records existed at the time of this deposit and are cited with confirmed DOIs. No forward references to undeposited documents are included.

ID	Title	DOI
Theory paper	Domestication as Field-Specified Developmental Arrest	10.5281/zenodo.19094935
P7	Cross-Fostering / NCC Arrest Pre-Registration	10.5281/zenodo.19096128
P-INVERSION-1	Plant Architecture Inversion via Coherence Gradient Field Specification	10.5281/zenodo.19040399
LECA-A	LECA Resurrection Fungal Arm	10.5281/zenodo.18986790
FOXA1/EZH2	FOXA1/EZH2 Ratio as Universal Breast Cancer Subtype Classifier	10.5281/zenodo.18883922

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